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APPLIANCE WITH A CLEAR ICE MAKER AND ICE PRESS

Abstract

An appliance described herein can include an ice maker and a cooled storage compartment. The ice maker can include a cooling plate coupled to a mold cavity, where the cooling plate includes a cooling tube and an electric heater. The cooling tube can cool the mold cavity during an ice production mode in which an ice billet is formed in the mold cavity by spraying water from a nozzle into the mold cavity. The electric heater can heat the mold cavity during an ice harvesting mode in which the ice billet is released from the mold cavity and guided along a guide ramp to the cooled storage compartment, where the ice billet can remain until needed. The appliance can also include an ice press, which can have one or more interchangeable molds and electric heaters to melt an ice billet into a shape of the molds.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates generally to appliances and, more particularly, to an appliance that includes a clear ice maker and an ice press.

BACKGROUND

[0002] Appliances such as refrigerators, coolers, and other kitchen appliances can have built-in ice makers to produce ice. The ice makers are machines that convert water (e.g., tap water) into ice. The ice is normally formed as solid cubes of a prefixed shape, such as a rectangular or crescent shape. Often, the ice is visually cloudy due to impurities such as dissolved minerals and salts in the water used to make the ice. These impurities can significantly alter the taste of drinks containing the ice. The cloudiness of the ice may also be visually unpleasant to consumers.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIG. 1 shows an example of an appliance having an ice maker, a cooled storage container for ice billets, and an ice press according to some aspects of the present disclosure.

[0004] FIG. 2 shows an example of an ice maker according to some aspects of the present disclosure.

[0005] FIG. 3 shows an exploded view of an example of an ice maker according to some aspects of the present disclosure.

[0006] FIG. 4 shows an example of an ice chamber module of an ice maker according to some aspects of the present disclosure.

[0007] FIG. 5 shows an example of an ice maker and a cooled storage container according to some aspects of the present disclosure.

[0008] FIG. 6 shows an example of an ice press with interchangeable molds according to some aspects of the present disclosure.

[0009] FIG. 7 shows an example of a control system associated with an ice maker according to some aspects of the present disclosure.

[0010] FIG. 8 shows a flowchart of an example of an ice production mode according to some aspects of the present disclosure.

[0011] FIG. 9 shows a flowchart of an example of an ice storage mode according to some aspects of the present disclosure.

[0012] FIG. 10 shows a flowchart of an example of an ice harvesting mode according to some aspects of the present disclosure.

DETAILED DESCRIPTION

[0013] Certain aspects and features of the present disclosure relate to an automatic ice maker that can produce large, substantially clear ice pieces, referred to herein as ice billets. The ice maker can include a mold cavity having an open end and a closed end, where the closed end is formed by a

cooling plate. The ice maker can produce an ice billet by spraying water into the open end of the mold cavity to accumulate ice on the cooling plate, which is cooled by circulating refrigerant through a cooling tube embedded in the cooling plate. Once the ice billet is ready for harvesting, an electric heater embedded in the cooling plate can be activated to detach the ice billet from the cooling plate. The ice billet can drop by gravity onto a guide ramp, which can transfer the ice billet to a cooled storage compartment. The cooled storage compartment may be kept, for example, at -6°C . ($\pm 3^{\circ}\text{C}$).

[0014] Some examples may also involve an ice press that can be used to change a shape of an ice billet. A user can obtain an ice billet from the cooled storage compartment and manually load it into the ice press. The ice press can have one or more removable ice molds and one or more embedded electric heaters. When the ice press is turned on, the electric heaters may maintain the ice press and molds at a temperature of around 55°C ., which can reduce the risk of the user burning their fingers and expedite the ice molding process. The ice molds can be removable and interchangeable, to allow consumers to change and combine them according to their drinks or mood. The ice press can be designed to work with different ice molds that produce sizes and shapes of ice, such as spheres, cubes, diamonds, etc.

[0015] The ice maker, cooled storage compartment, and ice press can be attached to an appliance (e.g., a refrigerator appliance) to provide an all-in-one solution for creating, storing, and shaping ice for beverages or other products. The ice maker can be compact and readily interchangeable with conventional domestic ice makers that utilize direct cooling, making it easy to retrofit to existing appliances.

[0016] These illustrative examples are given to introduce the reader to the general subject matter discussed here and are not intended to limit the scope of the disclosed concepts. The following sections describe various additional features and examples with reference to the drawings in which like numerals indicate like elements but, like the illustrative examples, should not be used to limit the present disclosure.

[0017] FIG. 1 shows an example of an appliance **100** having an ice maker **106**, a cooled storage container **108** for ice billets **110**, and an ice press **104** according to some aspects of the present disclosure. The ice maker **106** can create the ice billets **110** and transfer them into the cooled storage container **108**, for example by sliding the ice billets **110** along a guide ramp that extends from the ice maker **106** to the cooled storage container **108**. The cooled storage container **108** can be kept at a temperature that is below freezing to prevent the ice billets **110** from melting.

[0018] The cooled storage container **108** may include one or more mechanisms to store, organize, and/or provide access to the ice billets **110**. One example of such a mechanism can include a carousel **112** of ice holders **114**. The ice holders **114** can include platforms, clamps, and/or other devices for storing the ice billets **110**. The ice maker **106** can deposit the ice billets **110** onto the ice holders **114**. The carousel **112** can then be rotated (e.g., manually or electrically) to present the ice billets **110** to a user. For instance, a user may press a button of the appliance **100** to operate an electric motor that rotates the carousel **112** around its central axis until an ice holder **114** containing an ice billet **110** is accessible to the user. In other examples, other types of mechanisms may be used to store, organize, and/or provide access to the ice billets **110**. For instance, the cooled storage container **108** can include angled storage shelves for storing the ice billets **110**. The ice maker can deposit newly formed ice billets to the backs of the shelves using a system of guide ramps. The ice billets can then slide along the angled shelves toward the front, for example as the ice billets in front are removed by the user.

[0019] The ice billets **110** can have a large domed shape that may be unsuitable for some types of drinks. So, the appliance **100** can include the ice press **104** for use in sculpting the ice billets **110** into a desired shape. The final ice shape can be selectively modified by removing and replacing one or more interchangeable molds of the ice press **104**. As shown, the ice press **104** can be separate from the ice maker **106**. The ice press **104** may be permanently integrated into the appliance **100** or

may be removable from the appliance **100** as desired. For instance, the appliance **100** may include a receptacle **116** for housing the ice press **104**. When using the ice press **104**, the user may maintain the ice press **104** in the receptacle **116** or may selectively remove the ice press **104** from the receptacle (e.g., to place it on a countertop), whichever is easier.

[0020] In this example, the appliance **100** also includes a water dispenser **102**. The water dispenser **102** can dispense water (e.g., tap water, filtered water, hot water, and/or carbonated water) into a cup of a user for drinking, cooking, etc. The water dispenser **102** can be separate from the ice maker **106** and the ice press **104**.

[0021] In addition to its ice making and storage capabilities, the appliance **100** may be designed to serve other purposes. For instance, the appliance **100** may be a domestic kitchen appliance that is primarily designed for a purpose other than ice making and, thus, its ability to make clear ice can be considered a secondary purpose. As one particular example, the appliance **100** may be a refrigerator with the primary purpose of storing food. Alternatively, the appliance **100** may have the primary purpose of making and storing ice.

[0022] FIG. 2 shows an example of an ice maker **106** according to some aspects of the present disclosure. As shown, the ice maker **106** includes an ice chamber module **202** that includes a cover plate **224** with evaporator fins **206**, side walls **222**, and a cooling plate **204**. The side walls **222** can enclose a mold cavity that may have a dome shape. The mold cavity can be the location in which an ice billet is formed. The mold cavity can be made of any suitable material, such as metal or plastic. The cooling plate **204** can form or be coupled to the top of the mold cavity. The cooling plate **204** may be made of any suitable metal material, such as aluminum or copper. The cover plate **224** can be affixed over top of the cooling plate **204** to help maintain the cooling plate **204** in position, among other things.

[0023] The cooling plate **204** can have a cooling tube **208** embedded therein. The cooling tube **208** is connected to the refrigerant circuit of the appliance to provide cooling capacity. The cooling tube **208** can be clamped between the cooling plate **204** and the cover plate **224** and fins **206**. The cover plate **224** and fins **206** can be cooled by direct contact with the cooling tube **208**. The fins **206** can serve as an evaporator to cool the isolated compartment air next to the mold cavity, keeping the ice from melting.

[0024] The cooling plate **204** can also have an electric heater **210** embedded therein. The electric heater **210** can be an electric defrost heater. The electric heater **210** is connected to an electrical circuit that is operable to heat the cooling plate **204**, which can melt the ice billet **212** and thereby detach the ice billet **212** formed in the mold cavity from the cooling plate **204**. The ice billet **212** can then drop, due to gravity, onto a guide ramp **214** that can extend to a separate storage container for storing the ice billet.

[0025] In order to make an ice billet **212**, water is pumped into a water tank **216** from the appliance. A pump **218** is used to direct water from the water tank **216** to a water outlet referred to herein as nozzle **220**, which is positioned below the mold cavity. The nozzle **220** sprays the water upwardly through the bottom opening of the mold cavity, so that ice accumulates in the mold cavity. The ice can accumulate in layers, for example starting on the cooling plate **204** and extending downward, until an ice billet **212** is formed. Excess water falls out of the mold cavity and back to the water tank **216**. This flow of water, and lack of pressurization, provide an even flow of water to the mold cavity. Ice accumulates on the cooling plate **204** while impurities are washed away, resulting in formation of a clear ice billet **212**.

[0026] Once a desired amount of ice has accumulated, the ice billet is harvested by switching off or by-passing a refrigerant valve from the refrigerant circuit and then activating the electric heater **210** to warm the cooling plate **204** until the ice billet **212** is released from the cooling plate **204**, and it drops by gravity out of the mold cavity onto the guide ramp **214**. The water left in the water tank **216** can then be discharged through a drainage system. The water tank **216** can be refilled by the appliance and the cycle can be repeated as necessary to create additional ice billets.

[0027] To help conserve water, in some examples the ice maker **106** can include a total dissolved solids (TDS) sensor to measure and monitor the concentration of impurities (e.g., salt) in the water tank **216**. In these examples, the water in the water tank **216** may only be discharged and replaced when the concentration of impurities in the water tank **216** meets or exceeds a predefined limit, which may be sufficiently high to impact the taste or appearance of the ice billets. For example, a processor can detect that the TDS level meets or exceeds the predetermined limit based on a measurement from a TDS sensor coupled to the water tank **216**. In response to such a detection, the processor can operate a discharge valve to direct water to a drain connected to the appliance instead of circulation through the nozzle **220**. By waiting until the TDS measurement meets or exceeds the predetermined limit to discharge the water from the water tank **216**, the same water may be reused for multiple ice billets until it begins to impact their taste or appearance. This may reduce the amount of water consumed by the ice maker **106**, as compared to using new water for each ice billet.

[0028] FIG. **3** shows an exploded view of an example of an ice maker **106** according to some aspects of the present disclosure. As shown, the ice maker **106** can include a cover plate **224** with evaporator fins **206**. The ice maker **106** can also include a cooling plate **204** with channels for receiving a cooling tube **208** and an electric heater **210**. The cooling plate **204** can be substantially square in shape and have the channels engraved into its top and bottom sides for receiving the cooling tube **208** and the electric heater **210**, respectively. The ice maker **106** can further include a conductive enclosure **300** for receiving an ice mold **302** defining a mold cavity. The conductive enclosure **300** may be die cast and made of any suitable thermally conductive metal material, such as aluminum or copper. The conductive enclosure **300** may be coupled to the cooling plate **204** to propagate thermal changes (e.g., heating and cooling) originating from the cooling plate **204** to the side walls of the ice mold **302**, which may help in the creation or removal of an ice billet. The conductive enclosure **300** can be enclosed in a protective casing made from the side walls **222a-b** and positioned above a housing **304**. The guide ramp **214** can be positioned in the housing **304** to catch a falling ice billet (from the ice mold **302**) and direct it to a cooled storage container. The nozzle **220** and pump **218** can be used to direct water through the bottom opening of the ice mold **302** into the mold cavity to create an ice billet. The nozzle **220** can be positioned below the guide ramp **214** and oriented upwards to spray water through perforations in the guide ramp **214** to create the ice billet.

[0029] FIG. **4** shows an example of an ice chamber module **202** of an ice maker according to some aspects of the present disclosure. As shown, the ice chamber module **202** can include a conductive enclosure **300** coupled to a cooling plate **204** that contains an electric heater **210** and a cooling tube **208**. A cover plate **224** with evaporator fins **206** can be coupled over top of the cooling plate **204**. The conductive enclosure **300** can include an ice mold **302** with a mold cavity **402** into which water can be sprayed to create an ice billet having a shape defined by the mold cavity **402**.

[0030] FIG. **5** shows an example of an ice maker **106** and a cooled storage container **108** according to some aspects of the present disclosure. As shown, a drain assembly **502** can be affixed below the fins **206** of the ice maker **106** to direct water created from defrosting of the fins **206** to the water tank **216** or elsewhere. A duct **504** can also be positioned around the fins **206**. Air can be circulated through the duct **504** by a fan **506**. This can help cool the storage container **108** and thus facilitate storage of produced ice billets **212**.

[0031] As noted earlier, it may be desirable to reshape the ice billets. To that end, an ice press may be used in some examples. FIG. **6** shows an example of such an ice press **600**. The ice press **600** can be manually operated by a user independently of the ice maker **106**.

[0032] The ice press **600** can include a pair of molds disposed opposite to each other in mold housings **602**, **604**. The pair of mold housings **602**, **604** can be vertically openable and closable relative to one another. The upper mold housing **602** may contain an upper mold (not shown) and the lower mold housing **604** may contain a lower mold **610**. The upper mold and upper mold

housing **602** may be vertically movable together with respect to the lower mold and lower mold housing **604**, or vice versa, along one or more guide rails **606**. For example, by manually releasing the handle **608**, the upper mold housing **602** containing the upper mold can fall downwardly along one or more guide rails **606** toward the lower mold **610** in the lower mold housing **604**.

[0033] To operate the ice press **600**, an ice billet can be disposed between the upper and lower molds. The upper mold can be lowered onto the ice billet. The ice billet can be melted via one or more electric heaters **616a-b** in the upper mold housing **602** and/or the lower mold housing **604**. The continued application of force and heat can reshape the ice billet into a shape defined by the upper and lower molds.

[0034] The upper and lower molds can be interchangeable molds that can be selectively removed from the upper and lower mold housings **602**, **604**, respectively, and replaced with different interchangeable molds **612a-b**. For example, a user can manually detach the lower mold **610** from the lower mold housing **604** and replace the lower mold **610** with a different mold **612a**, which can result in a different ice shape. The interchangeable molds can be sized to fit into corresponding receiving areas of the mold housings **602**, **604** for use in the ice press **600**.

[0035] FIG. 7 shows an example of a control system **700** associated with an ice maker **106** according to some aspects of the present disclosure. The control system **700** includes a processor **702** coupled to a memory **704**. The processor **702** can include one processing device or multiple processing devices. Examples of the processor **702** include a Field-Programmable Gate Array (FPGA), an application-specific integrated circuit (ASIC), and a microprocessor. The processor **702** can execute instructions **706** stored in the memory **704** to perform one or more operations. In some examples, the instructions **706** can include processor-specific instructions generated by a compiler or an interpreter from code written in any suitable computer-programming language, such as C, C++, C#, and Java.

[0036] The memory **704** can include one memory device or multiple memory devices. The memory **704** can be volatile or non-volatile (i.e., the memory **704** can retain stored information when powered off). Examples of the memory **704** include electrically erasable and programmable read-only memory (EEPROM), flash memory, or any other type of non-volatile memory. At least a portion of the memory device includes a non-transitory computer-readable medium. A computer-readable medium can include electronic, optical, magnetic, or other storage devices capable of providing the processor **702** with the instructions **706** or other program code. Examples of a computer-readable medium include magnetic disks, memory chips, ROM, random-access memory (RAM), an ASIC, a configured processor, optical storage, or any other medium from which a computer processor can read the instructions **706**.

[0037] In some examples, the processor **702** can be coupled to a valve **708** of a refrigerant circuit **707**. As described above, the processor **702** can operate the valve **708** to circulate refrigerant through the cooling tube to cool a mold cavity of the ice maker. The processor may also be coupled to an electrical circuit **710**, which may for example include one or more switching components (e.g., transistors or relays). The processor **702** can operate the electrical circuit **710** to activate the electric heater **210**, which can be used to remove an ice billet from the mold cavity.

[0038] In some examples, the processor **702** can be coupled to one or more load sensors **712** (e.g., weight or force sensors). The processor **702** can receive load measurements from the load sensor **712**. The load measurements can indicate an amount of water in the water tank of the ice maker. The processor **702** may use the load measurement to determine how much water is present in the water tank (e.g., whether the water tank is full, half full, or empty). The processor **702** may use this information to inform other decisions, such as whether to begin or end an ice production mode. For instance, to make an ice billet, the processor **702** may initiate the ice production mode. During the ice production mode, the processor **702** can operate a fill system **716** to pump water into the water tank of the ice maker. The processor **702** can operate the fill system **716** until the processor **702** receives a load measurement from the load sensor **712** indicating that the load meets or exceeds a

first predefined threshold. The first predefined threshold may correspond to a sufficient amount of water to create an ice billet. In response to determining that the load measurement meets or exceeds the first predefined threshold, the processor **702** may deactivate the fill system **716**.

[0039] Next, the processor **702** may activate a pump **218** of the ice maker to begin spraying water into the ice mold to create the ice. During this process, the processor **702** can continue to monitor the load associated with the water in water tank. If the load falls below a second predefined threshold, it may mean that a sufficient amount of water has been converted into an ice billet in the ice mold. So, the processor **702** can deactivate the pump **218** to end the ice production mode. The processor **702** may also initiate an ice harvesting mode.

[0040] In the ice harvesting mode, the processor **702** can activate the electric heater **210** (e.g., by transmitting a control signal to the electrical circuit). This can detach the ice billet from the ice mold. The ice billet can fall onto the guide ramp and slide into a cooled storage container.

[0041] In some examples, the processor **702** can be coupled to one or more transfer sensors **714** (e.g., optical sensors, load sensors, ultrasonic transducers, etc.). The transfer sensors can be positioned on the guide ramp or in the ice storage container to detect the transfer of the ice billet to the storage container and transmit corresponding sensor signals to the processor **702**. In response to receiving the sensor signals, the processor **702** may activate a drainage system **718** to remove any excess water from the water tank. The drainage system **718** can include one or more valves that may be opened to drain the excess water from the water tank. Alternatively, the processor **702** may automatically activate the drainage system **718** at the conclusion of the ice production mode, at the conclusion of the ice harvesting mode, or at another point in time, irrespective of a sensor signal from the transfer sensor **714**.

[0042] In some examples, the processor **702** may activate the drainage system **718** in response to determining that a level of impurities (e.g., salt) in the water tank meets or exceeds a predefined threshold. For instance, the processor **702** can use a TDS sensor **720** positioned in the water tank to monitor the TDS level of the water in the tank. In response to determining that the TDS level meets or exceeds a predetermined limit, the processor can operate drainage system **718** to expel water from the water tank. By waiting until the TDS measurement meets or exceeds the predetermined limit to discharge the water from the water tank, the same water may be reused for multiple ice billets until it begins to impact their taste or appearance. This may reduce the amount of water that is wasted by the ice maker.

[0043] FIGS. **8-10** show flowcharts of examples of different operating modes associated with a control system according to some aspects of the present disclosure. Other examples may involve more, fewer, different, or difference sequences of steps than are shown in these figures. FIGS. **8-10** are described below with reference to the components of FIG. **7** described above.

[0044] Referring now to FIG. **8**, shown is an example of an ice production mode according to some aspects of the present disclosure. The ice production mode can begin at block **802**, where the processor **702** can determine whether the ice maker is turned on. If not, the system can enter an ice storage mode. Otherwise, the process can proceed to block **804**.

[0045] At block **804**, the processor **702** turns on a refrigerant valve and a compressor. The refrigerant valve and the compressor may be parts of the refrigerant circuit **707** of FIG. **7**. For example, the refrigerant valve may be the valve **708** of FIG. **7**. Turning on the refrigerant valve and the compressor can begin to circulate refrigerant to the cooling tube of the cooling plate.

[0046] At block **806**, the processor **702** determines whether the storage container (e.g., storage container **108** of FIG. **1**) for the ice billets is full. For instance, the processor **702** may receive sensor signals from one or more fill sensors associated with the storage container, where the sensor signals indicate whether the storage container is full. An example of such a fill sensor can be an optical sensor. Based on the sensor signals, the processor **702** can determine whether the storage container is full. If the processor **702** determines that the storage container is full, the system can enter an ice storage mode. Otherwise, the process can continue to block **808**.

[0047] At block **808**, the processor **702** determines whether a temperature of the storage container is less than or equal to a first predefined threshold, such as -5°C . To do so, the processor **702** can receive a temperature measurement from a temperature sensor associated with (e.g., positioned in) the storage container. If the temperature of the storage container is greater than the first predefined threshold, the process can return to block **804** and iterate. This step can help ensure that the storage container is sufficiently cold to handle the ice billets. If the temperature of the storage container is less than or equal to the first predefined threshold, the process can continue to block **810**.

[0048] At block **810**, the processor **702** determines whether a temperature of the cooling plate (e.g., cooling plate **204** of FIG. 3) is less than or equal to a second predefined threshold, such as -20°C . To do so, the processor **702** can receive a temperature measurement from a temperature sensor associated with (e.g., positioned on) the cooling plate. If the temperature of the cooling plate is greater than the second predefined threshold, the process can return to block **804** and iterate. This step can help ensure that the cooling plate is sufficiently cold to create the ice billets. If the temperature of the cooling plate is less than or equal to the second predefined threshold, the process can continue to block **812**.

[0049] At block **812**, the processor **702** activates (e.g., opens) a water valve to fill a water tank of the ice maker. The water valve may be part of the fill system **716** of FIG. 7.

[0050] At block **814**, the processor **702** determines whether a load measurement associated with the water tank is greater than or equal to a third predefined threshold. To do so, the processor **702** can receive a load measurement from a load sensor **712** associated with the water tank. If the load measurement is less than the third predefined threshold, the process can return to block **814** and iterate. This step can help ensure that there is sufficient water in the tank to create an ice billet. If the load measurement is greater than or equal to the third predefined threshold, the process can continue to block **816**.

[0051] At block **816**, the processor **702** deactivates (e.g., closes) the water valve to stop water flow into the water tank of the ice maker.

[0052] At block **818**, the processor **702** activates the water pump (e.g., pump **218** of FIG. 3) of the ice maker to spray water into the ice mold to create an ice billet.

[0053] During this spraying process, in which ice accumulates on the cooling plate to create the ice billet, the processor **702** can monitor the load measurements from the load sensor **712**. In some examples, the processor **702** can reduce the water flow over time based on the load measurements to prevent the water spray from perforating the ice billet. For example, the processor **702** can begin at 100% flow and gradually reduce the water flow over the course of building the ice billet. This can reduce the spray pressure over time, so that as the ice billet grows in size and gets closer to the spray nozzle, the pressure from the spray does not damage the ice billet. To determine how to reduce the water flow (e.g., spray pressure) over time, the processor **702** can apply a predefined algorithm. The predefined algorithm can depend on the load measurements. Decreasing load measurements can correspond to decreasing water levels in the tank, which in turn can correspond to an increase in the size of the ice billet. Thus, the load measurements can serve as an indicator of the amount of water left in the water tank, which in turn serves as a proxy for the amount of ice that has accumulated in the ice billet. Using these principles, the processor **702** can reduce the water flow based on the load measurements decreasing in value.

[0054] At block **820**, the processor **702** determines whether a load measurement from the load sensor **712** is less than or equal to a fourth predefined threshold. The fourth predefined threshold may correspond to the difference between the weight of the full water tank and the weight of an ice billet. For example, if an ice billet requires 50% of the water in the tank to create, then the fourth predefined threshold may correspond to 50% of the weight of a full water tank. As another example, if an ice billet requires 30% of the water in the tank to create, then the fourth predefined threshold may correspond to 70% of the weight of a full water tank. The water tank may be considered “full” when it has been filled to the level sufficient to deactivate the water valve in

blocks **814-816**. If the processor **702** determines that the load measurement is greater than the fourth predefined threshold, the process can remain at block **820**. This step can help ensure that an ice billet of sufficient size is created. If the processor **702** determines that the load measurement is less than or equal to the fourth predefined threshold, the process can continue to block **822**. [0055] At block **822**, the processor turns off the water pump. The process may then enter the ice harvesting mode.

[0056] Referring now to FIG. **9**, shown is an example of an ice storage mode according to some aspects of the present disclosure. In block **902**, the processor **702** turns on a refrigerant valve and a compressor. The refrigerant valve and the compressor may be parts of the refrigerant circuit **707** of FIG. **7**. For example, the refrigerant valve may be the valve **708** of FIG. **7**. Turning on the refrigerant valve and the compressor can begin to circulate refrigerant to the cooling tube of the cooling plate, which may also help to cool the storage container.

[0057] In block **904**, the processor **702** determines whether a temperature of the storage container is less than or equal to a first predefined threshold, such as -5° C. To do so, the processor **702** can receive a temperature measurement from a temperature sensor associated with (e.g., positioned in) the storage container. If the temperature of the storage container is greater than the first predefined threshold, the process can return to block **902** and iterate. This step can help ensure that the storage container is sufficiently cold to handle the ice billets. If the temperature of the storage container is less than or equal to the first predefined threshold, the process can continue to block **906**.

[0058] In block **906**, the processor **702** determines whether the storage container is full. For example, the processor **702** may receive sensor signals from one or more fill sensors associated with the storage container, where the sensor signals indicate whether the storage container is full. Based on the sensor signals, the processor **702** can determine whether the storage container is full. If the processor **702** determines that the storage container is full, the system can return to block **902**. Otherwise, the system can enter the ice making mode.

[0059] Referring now to FIG. **10**, shown is an example of an ice harvesting mode according to some aspects of the present disclosure. In block **1002**, the processor **702** determines whether a temperature of the storage container is less than or equal to a first predefined threshold, such as -5° C. To do so, the processor **702** can receive a temperature measurement from a temperature sensor associated with (e.g., positioned in) the storage container. If the temperature of the storage container is greater than the first predefined threshold, the process can return to block **1002** and iterate. This step can help ensure that the storage container is sufficiently cold to handle the ice billets. If the temperature of the storage container is less than or equal to the first predefined threshold, the process can continue to block **1004**.

[0060] At block **1004**, the processor **702** turns off a refrigerant valve and a compressor. The refrigerant valve and the compressor may be parts of the refrigerant circuit **707** of FIG. **7**. For example, the refrigerant valve may be the valve **708** of FIG. **7**. Turning off the refrigerant valve and the compressor can prevent refrigerant from circulating to the cooling tube of the cooling plate.

[0061] At block **1006**, the processor **702** turns on the electric heater **210** associated with the cooling plate. For example, the processor **702** can transmit a control signal to the electrical circuit **710** of FIG. **7** to turn on the electric heater **210**.

[0062] At block **1008**, the processor **702** determines whether a temperature of the cooling plate is greater than or equal to a fifth predefined threshold, such as 25° C. To do so, the processor **702** can receive a temperature measurement from a temperature sensor associated with (e.g., positioned in) the cooling plate. If the temperature of the cooling plate is less than or equal to the cooling plate predefined threshold, the process can return to block **1006** and iterate. This step can help ensure that the cooling plate is sufficiently warm to detach the ice billet. If the temperature of the cooling plate is greater than or equal to the fifth predefined threshold, the process can continue to block **1010**.

[0063] At block **1010**, the processor **702** turns off the electric heater **210** associated with the

cooling plate.

[0064] At block **1012**, the processor **702** determines whether the ice billet transferred to the storage container. For example, the processor **702** can detect a sensor signal from a transfer sensors **714**. The sensor can be positioned in relation to the storage container to detect the transfer of the ice billet. Based on detecting the sensor signal, the processor **702** can determine that the ice billet transferred to the storage container.

[0065] If the processor **702** determines that the ice billet was transferred to the storage container, the process can continue to block **1014**. Otherwise, the process can return to block **1012** and iterate.

[0066] At block **1014**, the processor **702** activates a drainage system **718** to drain the water tank of the ice maker. For example, the processor **702** can transmit a control signal to a drainage valve of the drainage system **718** to open the drainage valve, so that water in the water tank can be drained from the water tank.

[0067] The foregoing description of certain examples, including illustrated examples, has been presented only for the purpose of illustration and description and is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Numerous modifications, adaptations, and uses thereof will be apparent to those skilled in the art without departing from the scope of the disclosure. For instance, any examples described herein can be combined with any other examples to yield further examples.

Claims

1. An appliance comprising: an ice press including: an interchangeable mold; a first electric heater configured to heat ice positioned in the interchangeable mold to melt the ice into a shape of the interchangeable mold; and a guide rail along which the interchangeable mold is vertically movable to mold the ice into the shape of the interchangeable mold; an ice maker including: an ice mold defining a mold cavity; a cooling plate coupled to the ice mold, the cooling plate including a cooling tube and a second electric heater, the cooling tube being configured to cool the mold cavity and the second electric heater being configured to heat the mold cavity; a guide ramp extending from below the mold cavity to a cooled storage container, the guide ramp being configured to transfer an ice billet created in the ice mold to the cooled storage container; and the cooled storage container for storing one or more ice billets.
2. The appliance of claim 1, wherein the ice press includes a first interchangeable mold and a second interchangeable mold, the first interchangeable mold being movable along one or more guide rails toward the second interchangeable mold to shape the ice, and wherein the ice press is configured to be manually operated by a user independently of the ice maker.
3. The appliance of claim 1, wherein the ice press is selectively removable from the appliance, and wherein the appliance includes a receptacle for storing the ice press.
4. The appliance of claim 1, wherein the ice maker further comprises a water tank and a water pump, the water tank being positioned below a bottom opening in the mold cavity to capture water runoff during an ice production mode, and the water pump being configured to circulate water from the water tank through a nozzle into the mold cavity to produce the ice billet.
5. The appliance of claim 4, wherein the nozzle is positioned below the bottom opening of the mold cavity and oriented to spray the water upwards into the mold cavity.
6. The appliance of claim 5, wherein the guide ramp is positioned between the nozzle and the bottom opening of the mold cavity.
7. The appliance of claim 1, wherein the cooled storage container includes a carousel of ice holders, the carousel being rotatable about a central axis, and wherein the guide ramp is oriented to guide the ice billet from the ice maker to the carousel.
8. The appliance of claim 1, further comprising a water dispenser that is separate from the ice maker and the ice press, the water dispenser being usable to dispense water into a cup of a user.

- 9.** The appliance of claim 1, wherein the ice maker includes a cover plate with evaporator fins, the cover plate being disposed overtop of the cooling plate; and wherein the appliance further comprises a duct surrounding the evaporator fins, the duct extending from the evaporator fins to the cooled storage container, wherein the duct includes a fan configured to circulate cooled air to the cooled storage container.
- 10.** The appliance of claim 1, wherein the cooling plate is coupled to a top of the mold cavity, and further comprising: a refrigerant circuit coupled to the cooling tube, the refrigerant circuit being configured to circulate refrigerant through the cooling tube to cool the top of the mold cavity during an ice production mode; and an electrical circuit coupled to the second electric heater, the electrical circuit being configured to activate the second electric heater to heat the top of the mold cavity during an ice harvesting mode.
- 11.** The appliance of claim 1, further comprising a load sensor coupled to a processor, the processor being configured to initiate an ice harvesting mode based on a load measurement from the load sensor being below a threshold level.
- 12.** The appliance of claim 1, further comprising a transfer sensor coupled to a processor, the processor being configured to: receive a sensor signal from the transfer sensor, the sensor signal indicating that the ice billet was transferred from the ice maker to the cooled storage container; and based on receiving the sensor signal, drain excess water from a water tank by operate a valve associated with the water tank, wherein the water tank serves as a source of water sprayed through a nozzle during an ice production mode.
- 13.** A method comprising: cooling, by an ice maker of an appliance, a mold cavity by circulating refrigerant through a cooling tube in a cooling plate coupled to the mold cavity; creating, by the ice maker, an ice billet in the mold cavity by spraying water from a nozzle below the mold cavity through a bottom opening of the mold cavity during an ice production mode; releasing, by the ice maker, the ice billet from the mold cavity using an electric heater in the cooling plate during an ice harvesting mode, wherein the ice harvesting mode is subsequent to the ice production mode, and wherein the electric heater is separate from the cooling tube; and guiding, by the ice maker, the ice billet along a guide ramp to a cooled storage container.
- 14.** The method of claim 13, further comprising: shaping, by an ice press that is separate from the ice maker, the ice billet into a shape defined by an interchangeable mold of the ice press, wherein the ice billet is shaped at least in part by applying heat from another electric heater of the ice press to the ice billet in the interchangeable mold.
- 15.** The method of claim 13, wherein the appliance includes a receptacle for storing an ice press, the ice press being selectively removable from the appliance and usable to shape the ice billet.
- 16.** The method of claim 13, wherein the cooled storage container includes a carousel of ice holders, and wherein the guide ramp is oriented to guide the ice billet from the ice maker to the carousel.
- 17.** The method of claim 13, wherein the cooled storage container includes one or more ramps for storing ice billets, each of the one or more ramps being oriented at a downward angle from a back of the cooled storage container to a front of the cooled storage container.
- 18.** The method of claim 13, further comprising: operating a refrigerant circuit coupled to the cooling tube to circulate the refrigerant through the cooling tube to cool the mold cavity during the ice production mode; and operating an electrical circuit coupled to the electric heater to heat the mold cavity during the ice harvesting mode.
- 19.** The method of claim 13, wherein the water for creating the ice billet is stored in a water tank, and further comprising: receiving, by a processor, a load measurement from a load sensor coupled to the water tank; and initiating, by the processor, the ice harvesting mode based on the load measurement.
- 20.** An ice maker, comprising: an ice mold defining a mold cavity; a nozzle positioned below a bottom opening of the mold cavity and oriented to spray water into the mold cavity; a water tank; a

cooling plate coupled to a top of the mold cavity, the cooling plate including a cooling tube and an electric heater, the cooling tube being separate from the electric heater; a processor; and a memory storing instructions that are executable by the processor for causing the processor to: initiate an ice production mode in which an ice billet is formed in the mold cavity by spraying the water from the water tank through the nozzle into the mold cavity to accumulate ice on the cooling plate, wherein initiating the ice production mode involves operating a refrigerant circuit to direct refrigerant through the cooling tube; and initiate an ice harvesting mode in which the ice billet is released from the cooling plate, wherein initiating the ice harvesting mode involves operating an electrical circuit to activate the electric heater.
